

# Exercises tmap course

## tmap

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1. Load the library **tmap**
2. Take a look at the dataset `World` (included in **tmap**). Also open the documentation page `?World`.
3. Make the following World maps using `tm_shape()` and `tm_polygons()` :
  - a. Fill all country areas with orange. See [vignette about visual variables](#) for some examples
  - b. Fill all country areas with colors that show a data variable (a variable from `World`)
  - c. Change the color palette to a rainbow palette. See [vignette about scales](#).
4. Take a look at the dataset `metro`, also contained in **tmap**
5. Make a bubble map with `tm_shape()` and `tm_symbols()` where the size of each bubble is proportional to the number of inhabitants in 2020.
6. Combine the map created at step 3a with the last map.
7. Make an interactive version of this map. See [vignette about modes](#)

## cols4all

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9. Load the library **cols4all**
10. Run the dashboard with `c4a_gui()`
11. What color palette can you recommend for the following data variables in the `World` dataset: `"well_being"`, `"gdp_cap_est"`, and `"footprint"`?
12. Make a choropleth (see 3b) of each of these variables using the selected color palettes. (Tip: put a `-` prefix to reverse the colors)